QPCODE: ..... **Reg. No.:** .....
Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION MONTH 2026
FIFTH SEMESTER
MG5DSCBIF300 – PERL & BIOPERL PROGRAMMING
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks:** 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)
Students should attempt at least one question from each Course Outcome to enhance their overall outcome attainability.

Part A – Short Answer Questions

Answer any 5 questions. Each question carries 2 marks.

1. What are the main features of Perl? Mention any two advantages of Perl. [U][1]

2. Distinguish between scalar and array variables in Perl with one example each. [U][1]

3. What is the purpose of the foreach loop in Perl? [U][2]

4. What is pattern matching in Perl? Mention the role of anchors in regular expressions. [U][4]

5. What is a Perl subroutine? State how values can be passed to a subroutine. [U][6]

6. What is BioPerl? Mention any two applications of BioPerl in biological data processing. [U][8]

7. What is the Bio::Seq class used for in BioPerl? [U][8]

[5 × 2 = 10]

Part B – Short Essay Questions

Answer any 4 questions. Each question carries 5 marks.

8. Explain Perl data types and variables. Discuss scalars, arrays, numbers and strings with suitable examples. [U][1]

9. Explain conditional statements and looping constructs in Perl. Illustrate any two loops with examples. [A][2]

10. Explain Perl file handling. Describe creating, opening, reading, writing and closing files. [A][3]

11. Explain character sets, ranges, word boundaries and anchors used in Perl regular expressions with suitable examples. [U][4]

12. Explain how Perl regular expressions are used for finding all occurrences of a pattern and replacing a pattern. [A][5]

13. Explain built-in string handling functions and user-defined subroutines in Perl. Discuss passing values and returning values. [A][6]

[4 × 5 = 20]

Part C – Long Essay Questions

Answer any 2 questions. Each question carries 10 marks.

14. Explain Perl programming fundamentals in detail. Discuss data types, variables, operators, operator precedence, conditional statements, loops, input/output and file management with suitable examples. [A][1][2][3]

15. Explain Perl Regular Expressions and Functions in detail. Discuss pattern matching, word boundaries, anchors, character sets, case-insensitive matching, finding and replacing patterns, built-in functions, subroutines and Perl modules. [E][4][5][6][7]

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16. Explain BioPerl in detail. Discuss installation, general BioPerl classes, BioSeq, sequence input/output and manipulation, accessing GenBank, AlignIO and BLAST parsing, with suitable applications in Bioinformatics. [A][8]

17. Design a Perl/BioPerl-based solution for a biological sequence-processing problem. Explain the steps involved in reading sequence data, manipulating or parsing it, accessing relevant biological database information and producing useful output. [C][3][8]

[2 × 10 = 20]